

Brad Cai

(858) 429-8369 | bradcai24@gmail.com | [linkedin.com/in/bradcai](https://www.linkedin.com/in/bradcai) | bradleycai.github.io

PROJECTS

PBR in Custom Vulkan Engine | C++, Vulkan, GLSL

- Designed a physically based rendering system and asset pipeline in Vulkan.
- Implemented IBL with prefiltered radiance & irradiance environment maps and a precomputed BRDF integration LUT for the split sum approximation of the specular BRDF.
- Added logic for loading metallic-roughness, occlusion, and emissive data from glTF files to support PBR models.

Custom Vulkan Engine | C++, Vulkan, GLSL

- Built a modern real-time graphics engine using Vulkan 1.3, VMA, and C++ utilizing Dynamic Rendering and Buffer Device Address Vulkan features.
- Handled Vulkan CPU/GPU synchronization, command execution, and explicit GPU resource management of pipelines, descriptor sets, images, and buffers.
- Added the ability to load assets from glTF files and display them in the engine.
- Implemented a scene graph and node system for model hierarchies and organization.
- Built profiling tools and debugging overlays using Dear ImGui.

Snow Ground Shader | GLSL, GDShader, Godot

- Designed a real-time effect in the Godot game engine that simulates pressed snow from nearby objects.
- Used GLSL and GDShader in a compute and fragment shader to read camera depth values, process them into terrain height information, and display it in a visual effect of pressed snow using parallax mapping.

EXPERIENCE

Software Engineer

August 2019 – January 2025

ServiceNow

San Diego, CA

- Shipped, updated, and maintained 3 large customer facing products as a part of ServiceNow's core platform team.
- Led development and authored software architecture plans approved by the architecture review board.
- Optimized application performance by profiling hot paths and bottlenecks to reduce memory usage and runtime.
- Engineered core features and bug fixes using Java, SQL, and MariaDB within a large codebase.
- Built maintainable code using strong software design principles and software engineering practices.
- Collaborated in a Scrum team across six-month software development life cycles to deliver timely stable releases.
- Built DevOps scripts in Python to improve developer productivity and designed data visualization tools using Pandas, Plotly, and Jupyter to analyze customer usage metrics and drive development priorities.
- Implemented responsive front-end features using Javascript, AngularJS, and jQuery.
- Triaged issues, fixed defects, and authored tests to maintain a high level of product stability.

EDUCATION

University of California, Riverside

September 2015 – June 2019

Bachelor of Science in Computer Science

Riverside, CA

- Relevant Coursework: Computer Graphics (OpenGL, C++), GPU Computing & Programming (CUDA, C++), Graphics and Electronic Games (Unity, C#), Virtual Reality (Unity, C#), Concurrent Programming (C++), Applied Linear Algebra.

TECHNICAL SKILLS

Graphics APIs & Tools: Vulkan, OpenGL, RenderDoc

Languages (Graphics): C++, C, GLSL

Languages (Enterprise/Web): Java, SQL, Python, JavaScript

Tools & Engines: Godot, Unity, GNU/Linux, Git, CLI tools

Math: Linear algebra, vector math, trigonometry